

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I Sivadharshini.K (192524008) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Sivadharshini.K

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Smart Campus Subnet Design – 192.168.100.0/24

Overview

Starting block: **192.168.100.0/24** (256 total addresses, .0 to .255)

Subnets are allocated sequentially, from largest to smallest, so each block starts right after the previous one ends.

1–3. Network Details, Host Ranges & Usable Hosts

School of Computing – /25

Parameter	Value
Subnet Mask	255.255.255.128
Network Address	192.168.100.0
Broadcast Address	192.168.100.127
Valid Host Range	192.168.100.1 – 192.168.100.126
Usable Hosts	$2^7 - 2 = \mathbf{126}$

School of Electronics – /26

Parameter	Value
Subnet Mask	255.255.255.192
Network Address	192.168.100.128
Broadcast Address	192.168.100.191
Valid Host Range	192.168.100.129 – 192.168.100.190
Usable Hosts	$2^6 - 2 = \mathbf{62}$

School of Mechanical Engineering – /27

Parameter	Value
Subnet Mask	255.255.255.224
Network Address	192.168.100.192
Broadcast Address	192.168.100.223
Valid Host Range	192.168.100.193 – 192.168.100.222
Usable Hosts	$2^5 - 2 = \mathbf{30}$

Total usable hosts allocated: $126 + 62 + 30 = \mathbf{218}$

4. Remaining Unused IP Address Range

After allocating the three subnets, the address space from **192.168.100.224** through **192.168.100.255** is left unused.

Parameter	Value
Unused Range	192.168.100.224 - 192.168.100.255
Size	32 addresses
Equivalent Block	192.168.100.224/27 (available for future expansion, e.g., a Server Room, Admin Block, or Guest Wi-Fi subnet)

Summary Table

Department	CIDR	Network Address	Broadcast Address	Host Range	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.127	.1 - .126	126
School of Electronics	/26	192.168.100.128	192.168.100.191	.129 - .190	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223	.193 - .222	30
Unused/Reserved	/27	192.168.100.224	192.168.100.255	.225 - .254	30 (reserved for future use)

This design uses **Variable Length Subnet Masking (VLSM)** efficiently – larger departments get proportionally larger blocks, and the leftover /27 block keeps room for future growth (e.g., a new lab or administrative wing) without needing to renumber existing subnets.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.
3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

Given the IP block **10.10.0.0/16**, the available address range is:

- **Network Block:** 10.10.0.0 – 10.10.255.255
- **Total Addresses:** 65,536

The departments are allocated subnets from largest to smallest:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

1. Network Address and Broadcast Address

Department	Subnet Mask	Network Address	Broadcast Address
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191

Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223
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2. Valid Host IP Address Range

Department	Valid Host Range
Software Development	10.10.0.1 – 10.10.0.254
Quality Assurance	10.10.1.1 – 10.10.1.126
Human Resources	10.10.1.129 – 10.10.1.190
Executive Management	10.10.1.193 – 10.10.1.222

3. Number of Usable Hosts

Formula:

$$\text{Usable Hosts} = 2^{(32 - \text{Prefix Length})} - 2$$

Department	Prefix	Total Addresses	Usable Hosts
Software Development	/24	256	254
Quality Assurance	/25	128	126
Human Resources	/26	64	62
Executive Management	/27	32	30

4. Remaining Unused IP Address Range

Allocated subnets occupy:

- 10.10.0.0 – 10.10.0.255 (/24)
- 10.10.1.0 – 10.10.1.127 (/25)
- 10.10.1.128 – 10.10.1.191 (/26)
- 10.10.1.192 – 10.10.1.223 (/27)

The remaining addresses in the partially used 10.10.1.x block are:

- 10.10.1.224 – 10.10.1.255 (available as 10.10.1.224/27)

Additionally, the rest of the original /16 block is still unused:

- 10.10.2.0 – 10.10.255.255

Final Summary

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
Software Development	10.10.0.0/24	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192/27	10.10.1.193 – 10.10.1.222	10.10.1.223	30
Unused	10.10.1.224/27 and 10.10.2.0 – 10.10.255.255	10.10.1.225-10.10.1.254	10.10.1.225 for the /27	30 hosts in the remaining /27, plus the rest of the /16 block available for future subnetting

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**

3. Identify the network prefix using the /64 prefix length.
4. Calculate the total number of host addresses supported within the subnet.
5. the total number of possible host addresses available within this network.

Given IPv6 Network

2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert the given IPv6 address into its compressed notation

Rules:

- Remove leading zeros in each block.
- Replace consecutive blocks of zeros with :: (only once).

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the compressed address back into its full hexadecimal representation

Expanding 2001:db8:abcd::/64:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Identify the network prefix using the /64 prefix length

IPv6 address:

2001:0db8:abcd:0000 : 0000:0000:0000:0000

A /64 prefix means the first 64 bits (first four hexadecimal blocks) represent the network.

Network Prefix:

2001:0db8:abcd:0000::/64

or simply

2001:db8:abcd:0::/64

4. Calculate the total number of host addresses supported within the subnet

IPv6 address length = **128 bits**

Network prefix = **64 bits**

Host portion = $128 - 64 = 64$ bits

Number of host addresses:

$$2^{64} \\ = 18,446,744,073,709,551,616$$

Total host addresses supported = 18,446,744,073,709,551,616

5. Total number of possible host addresses available within this network

Since the host portion contains 64 bits, the number of unique host addresses is:

$$2^{64} \\ \text{Total possible host addresses available} = 18,446,744,073,709,551,616$$

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Given Routing Table:

Network Address	Prefix
192.168.1.0	/24
192.168.2.0	/24
192.168.3.0	/24

Destination IP Address: 192.168.2.45

1. Destination subnet to which the packet belongs

For a /24 subnet, the first three octets represent the network portion.

Destination IP: 192.168.2.45

Network Address:

192.168.2.0/24

Answer: The packet belongs to the subnet 192.168.2.0/24.

2. Matching routing table entry

Compare the destination network with the routing table:

- 192.168.1.0/24 ✗
- 192.168.2.0/24 ✓
- 192.168.3.0/24 ✗

Answer: The matching routing table entry is 192.168.2.0/24.

3. Next-hop network/interface used to forward the packet

Since the destination matches 192.168.2.0/24, the router forwards the packet through the interface (or next-hop) connected to the 192.168.2.0/24 network.

Answer: Forward through the interface connected to 192.168.2.0/24.

4. Default route if no match exists

If a destination IP does not match any routing table entry, the router uses the default route.

Default Route:

0.0.0.0/0

This route sends packets to the configured default gateway (next-hop router).

Answer: Use the default route 0.0.0.0/0.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

Given Information:

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Gateways:

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

A /26 subnet provides:

- 64 IP addresses per subnet
- 62 usable host addresses

1. Network Address and Broadcast Address

LAN	Network Address	Broadcast Address
Administration	192.168.10.0/26	192.168.10.63
Finance	192.168.10.64/26	192.168.10.127

2. Valid Host IP Address Range

LAN	Valid Host Range
Administration	192.168.10.1 – 192.168.10.62
Finance	192.168.10.65 – 192.168.10.126

3. Suitable IP Addresses for Three Computers in Each LAN

Administration LAN

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Finance LAN

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68

(Any unused addresses within the valid host range are acceptable.)

4. Default Gateway for Each LAN

LAN	Default Gateway
Administration	192.168.10.1

Finance	192.168.10.65
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Hosts must use these gateway addresses to communicate with devices outside their own subnet.